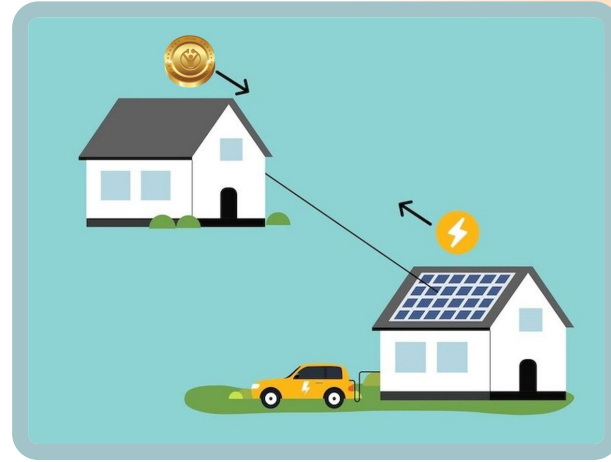




Blockchain-based Relayer Services

Batching & Submitting Multiple Energy Transactions in a Meta-Transaction

Supervisor: Assoc Prof. Gooi Hoay Beng

Presenter: Ma Siteng

Outline

01

Introduction

02

Literature Review

03

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04

Implementation

05

Conclusion

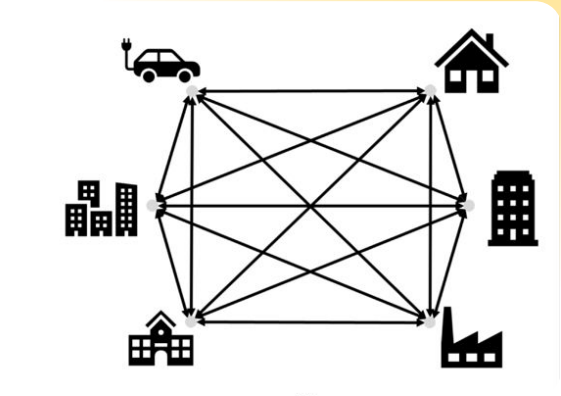
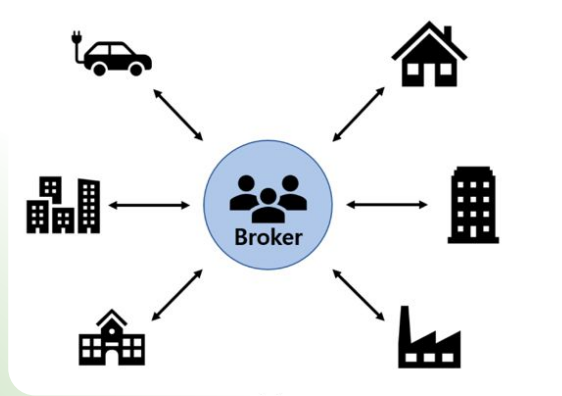


01

Introduction

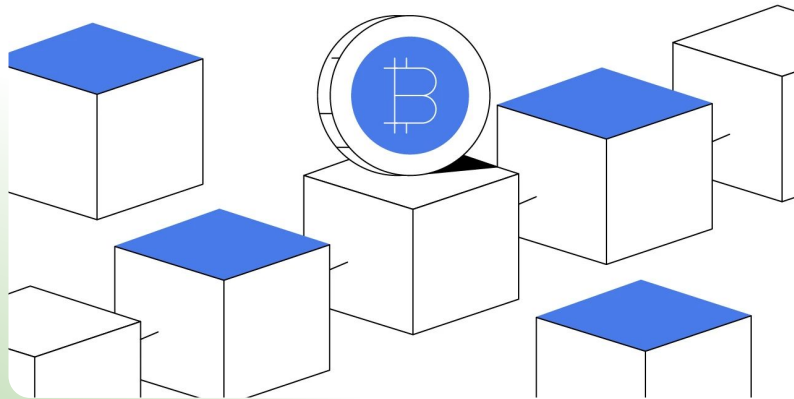
P2P Energy Trading

Prosumer-centric power trading system
where prosumers can sell the excess generated energy
directly to other prosumers with energy deficiency



Blockchain

Distributed ledger powered by consensus algorithm
able to record transactions with verifiability and
traceability, and offers programmable smart contract



Problem Statement

P2P energy trading and blockchain is a great combo, but there are still issues...



Efficiency



Usability



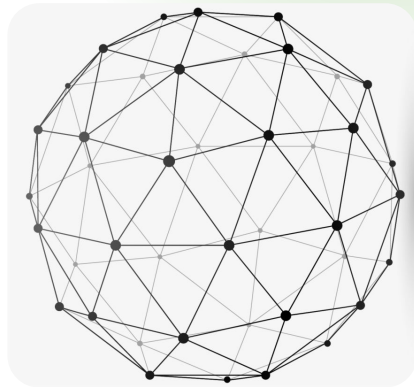
Efficiency

- Energy trading on blockchain is prone to congestion
- Limitation of blockchain architecture
- Transaction speed is limited in public blockchain

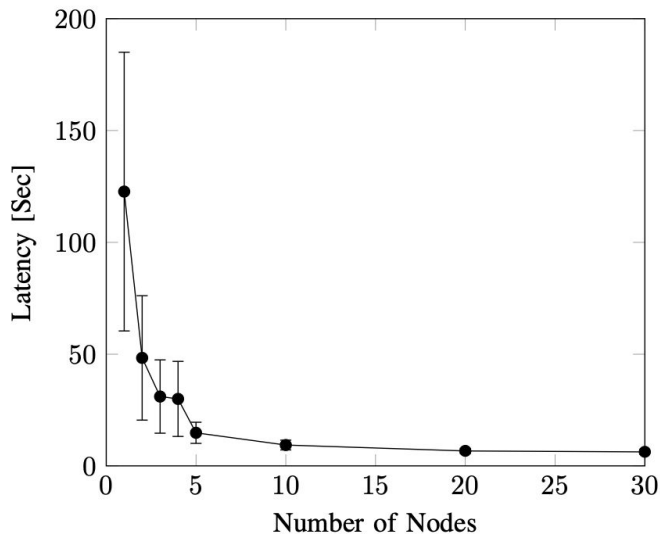
According to [data source](#), Ethereum handle **24** transactions / s

- Scale of network is limited in private or consortium blockchain

According to [research paper](#), blockchain latency increases as the number of nodes decreases



Public blockchain





Usability

- Gas fee is required to launch transaction on public blockchain
- Gas fee is paid in the blockchain's native token.
- Difficult for inexperienced prosumers
- Hinders the further adoption of blockchain

<https://foundation.app>

PLACE BID

◆ 0.1101

DETAILS

DATA

[EDIT](#)

foundation.app suggested \$53.05 **0.01785 ETH**
gas fee ⓘ

Likely in < 30 seconds

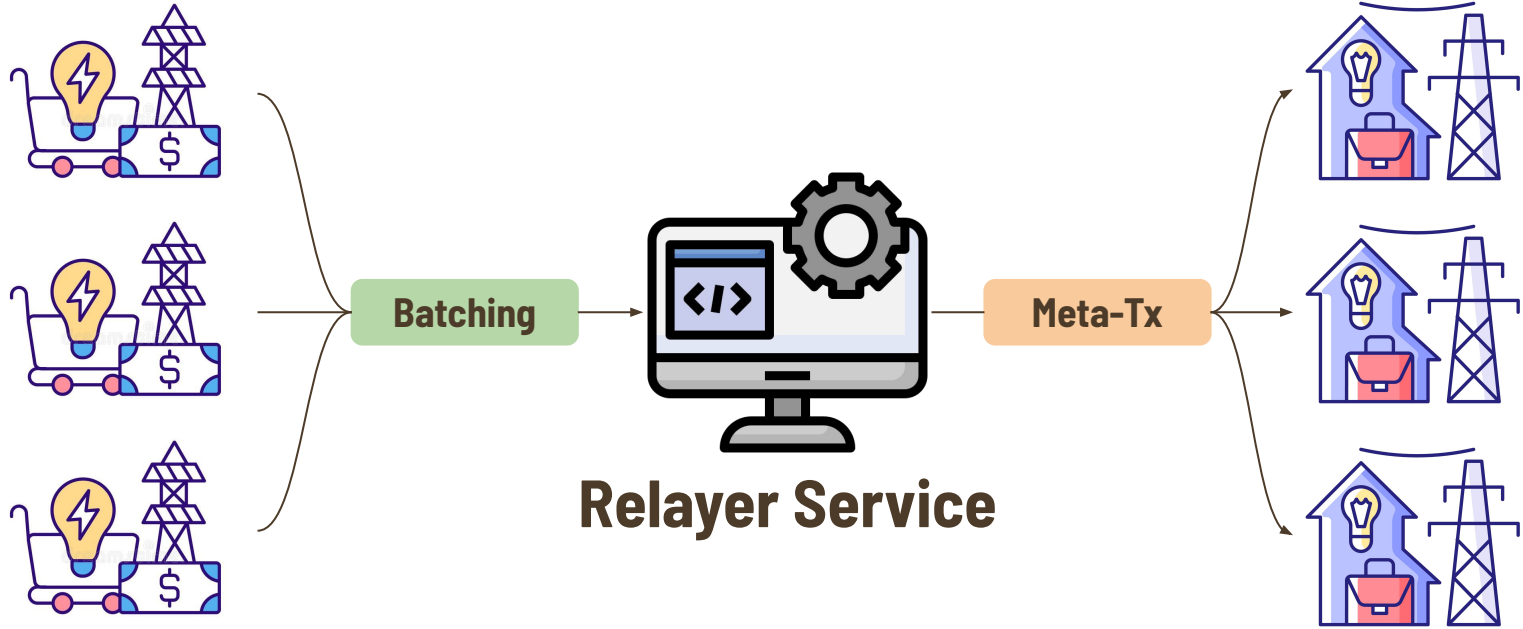
Max fee:
0.023172 ETH

Total \$380.29 **0.12795 ETH**

Amount + gas fee

Max amount: 0.133272 ETH

Objective



02

Literature Review



Major Concepts



P2P Energy Trading

- Definition
- Comparison
- Procedure
- Example



Blockchain

- Definition
- Ethereum
- Private vs Public
- Transaction



Smart Contract

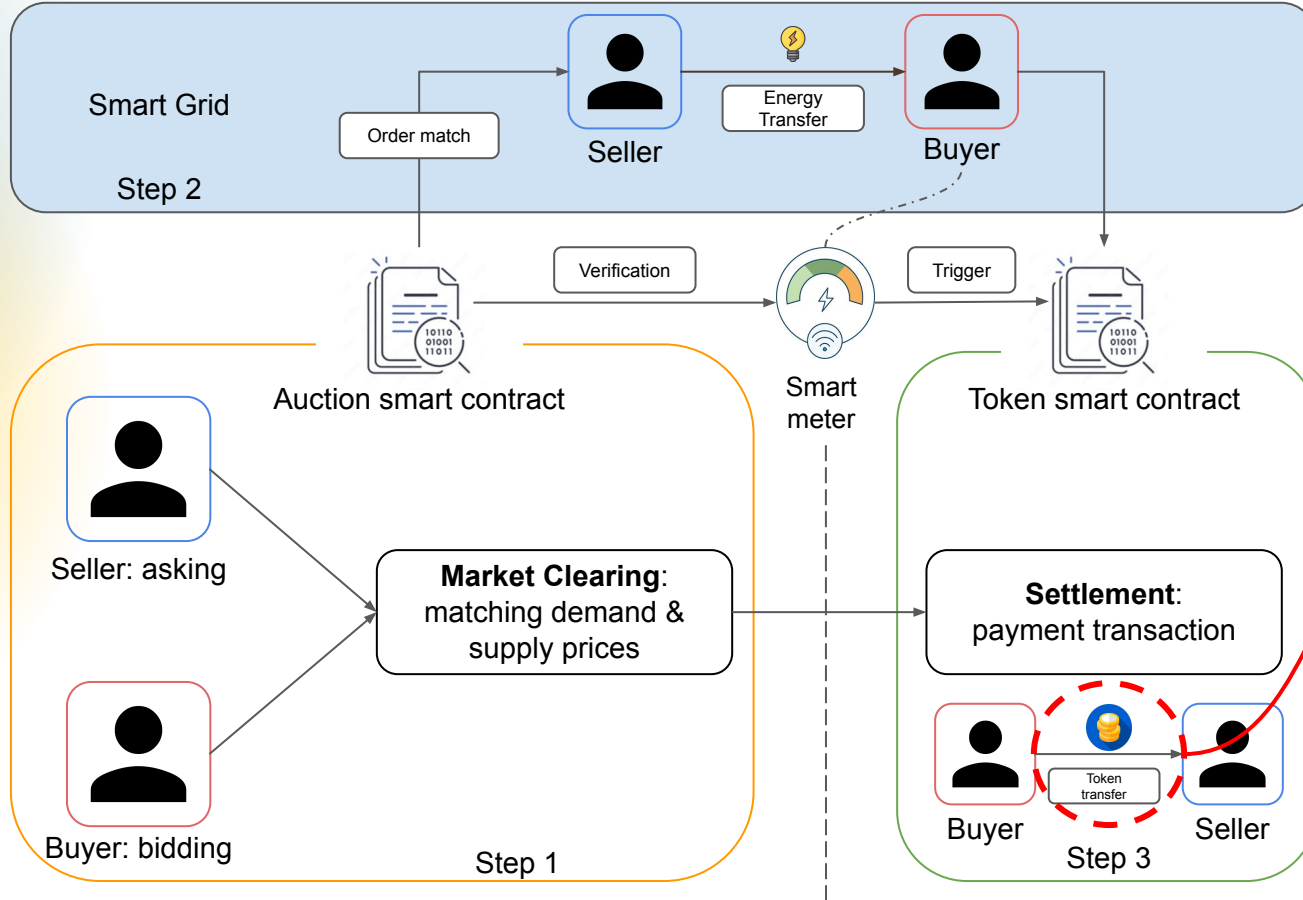
- Definition
- Programming
- Gas Fee



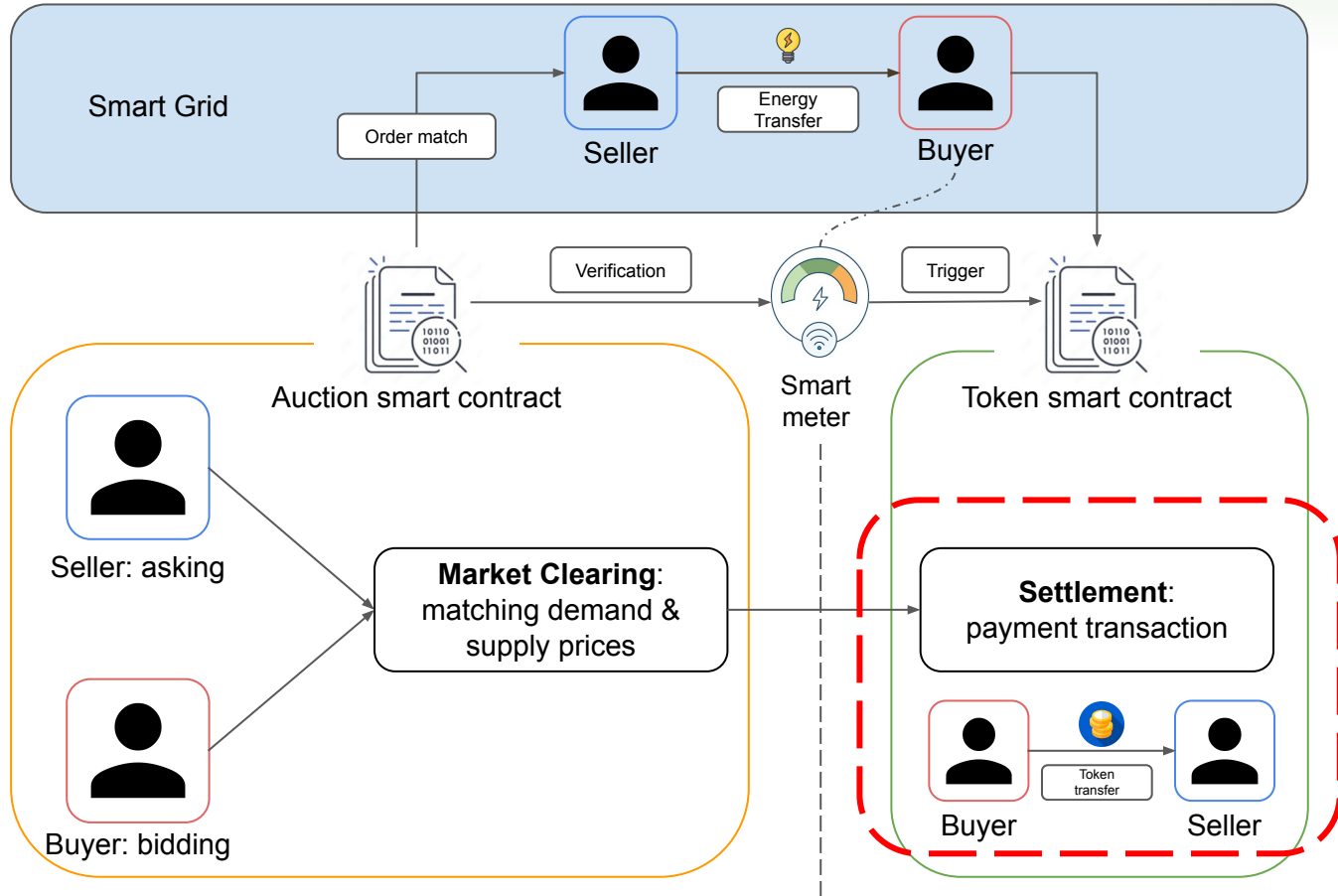
Digital Signature

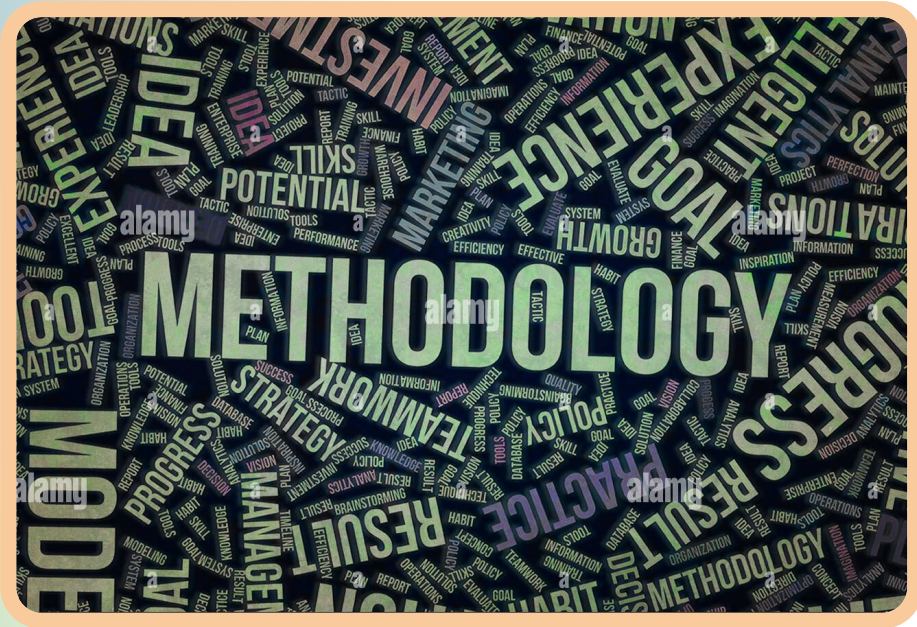
- Definition
- Public-key Cryptography
- Elliptic Curve Algorithm

Blockchain-based P2P Energy Trading



Scope of the Relay Service

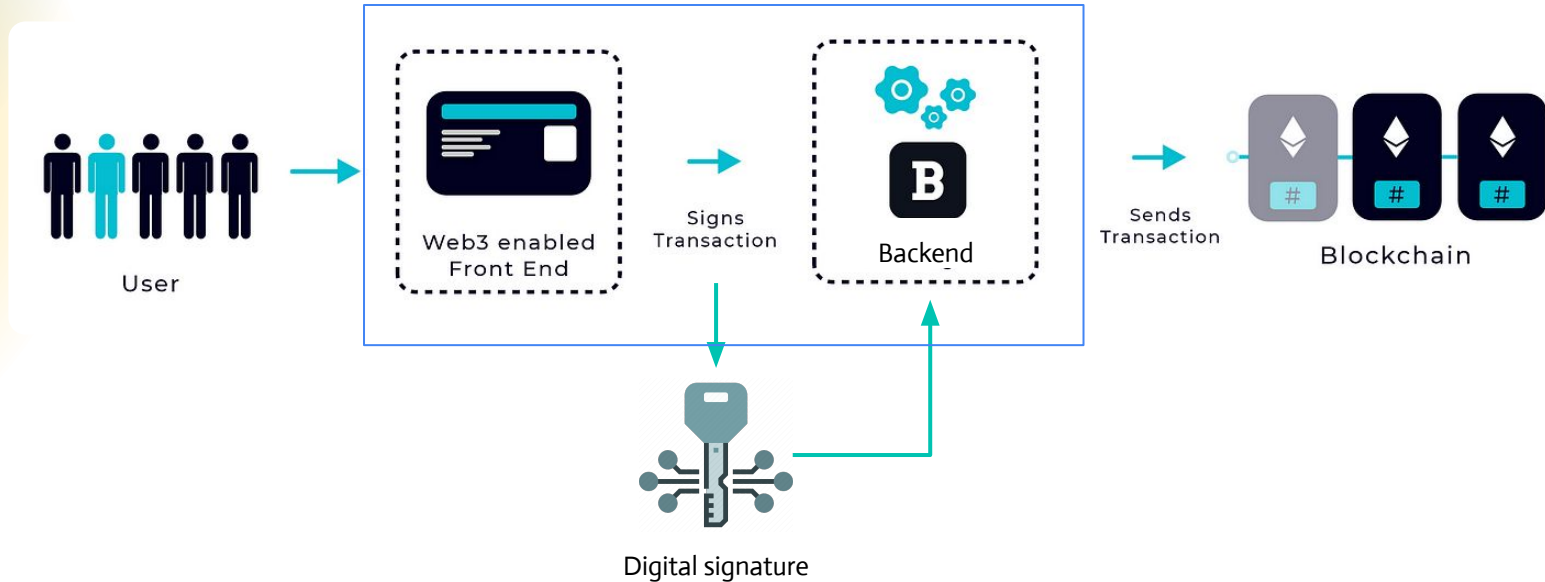




03 Methodology

Meta-Transaction

- A type of blockchain transaction that users do not need to pay gas fee to initiate
- Working process:



Batching Mechanism

- Multiple transaction requests are bundled into a single transaction
- Achieved at smart contract level
- Reduce number of one-to-one transactions, save blockchain node resources, reduce time

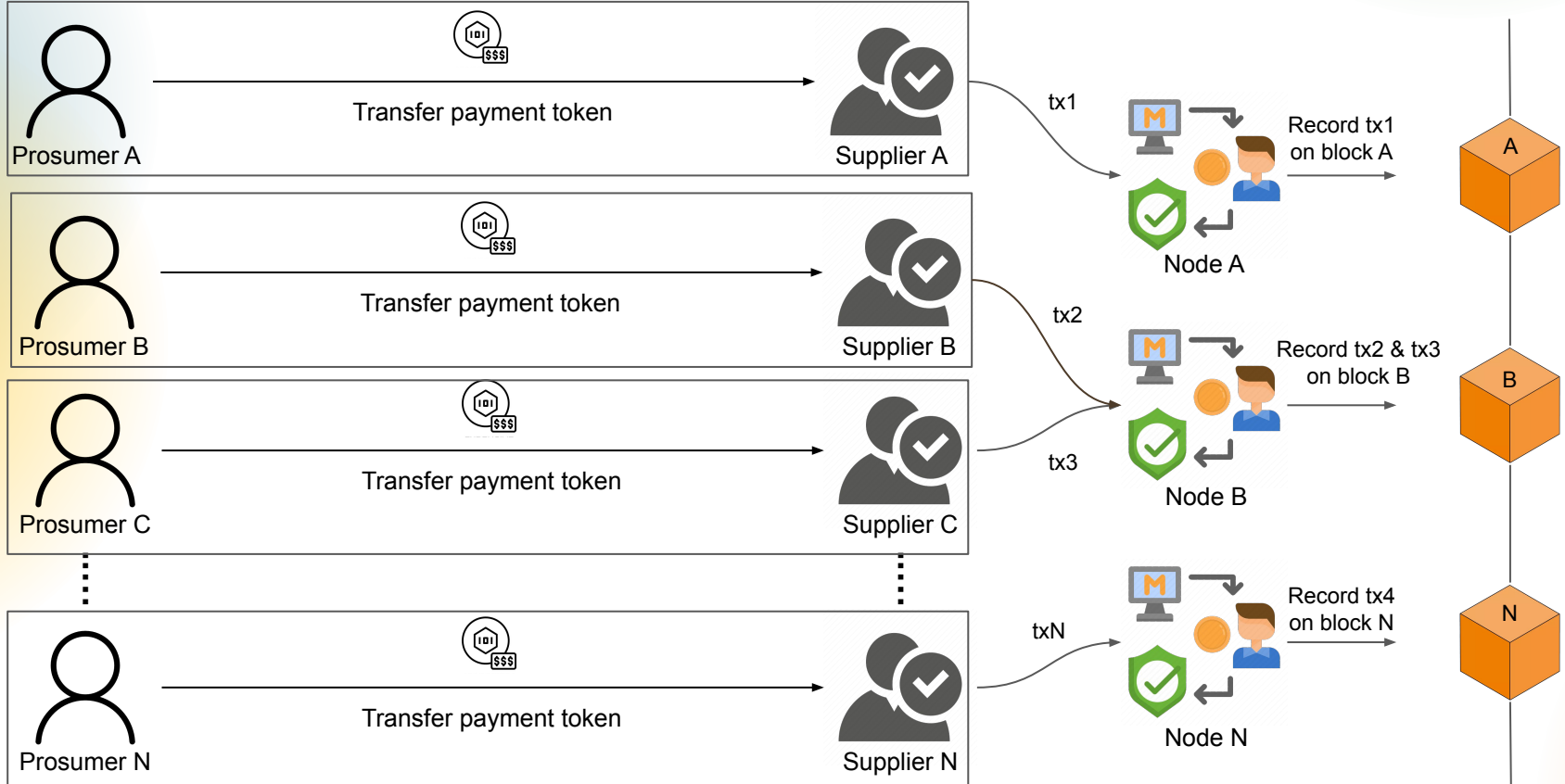


```

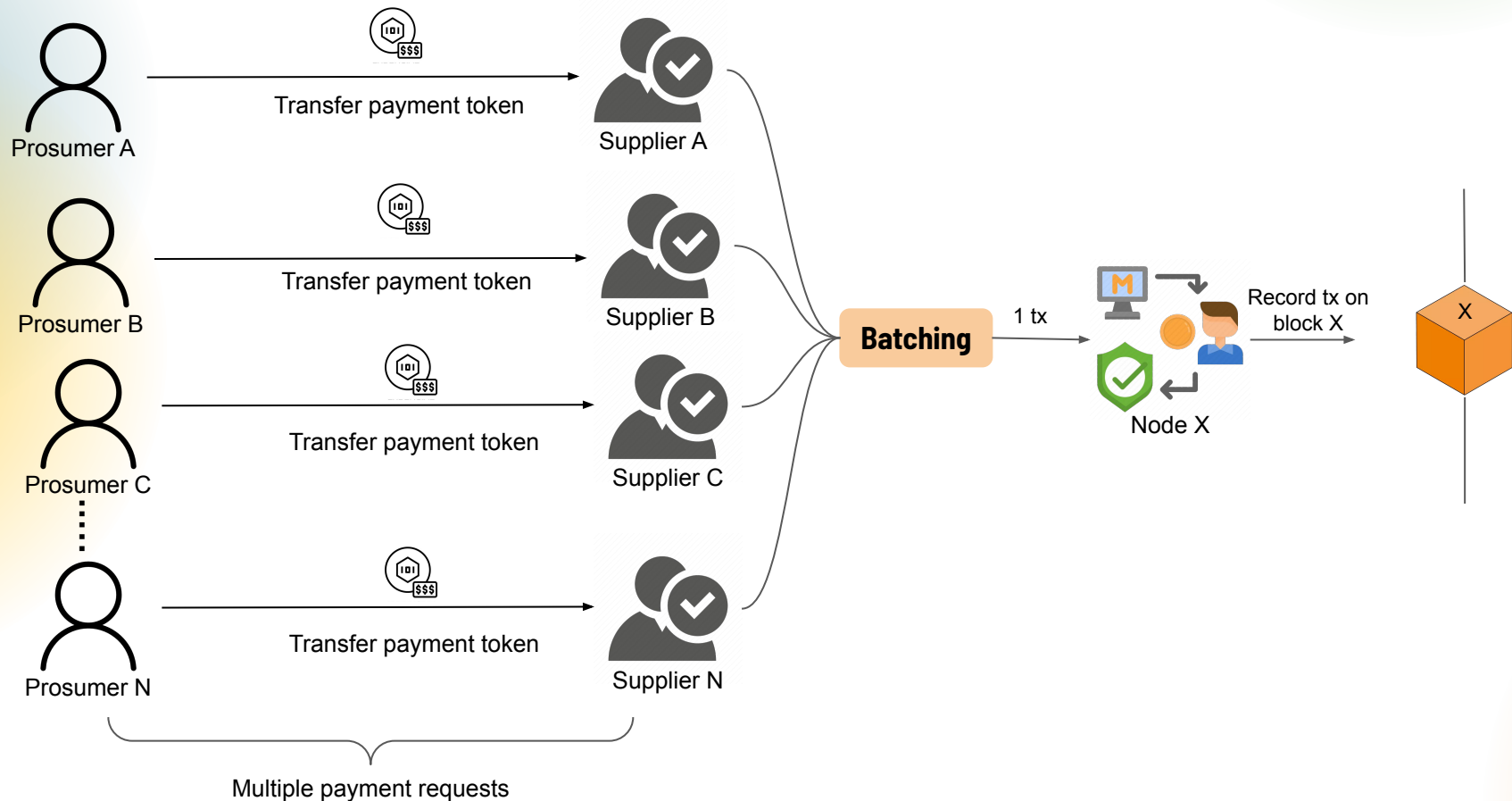
1 //src/standard_authority.sol
2
3 import "dsproxy/control/authority_sol";
4 import "dsproxy/control/auth_sol";
5
6 contract DSStandardAuthority is DSAuthority, DSAuth {
7     function DSStandardAuthority() {
8         _is_root[msg.sender] = true;
9     }
10
11     mapping(address->bool) public is_root;
12     mapping(address->mapping(address->mapping(bytes4->bool))) can_call;
13
14     function can_call(address caller
15         , address callee
16         , bytes4 sig )
17     constant
18         returns (bool)
19     {
20         return can_call[caller][callee][0x0000] == true
21             || can_call[caller][sig];
22     }
23
24     event set_can_call_event( address caller, address callee, bytes4 sig, bool can );
25
26     function set_can_call( address caller
27         , address callee
28         , bytes4 sig
29         , bool can )
30         auth{
31             returns (bool success)
32         {
33             _can_call[caller][sig] = can;
34             set_can_call_event(caller, callee, sig, can);
35             return true;
36         }
37     }

```

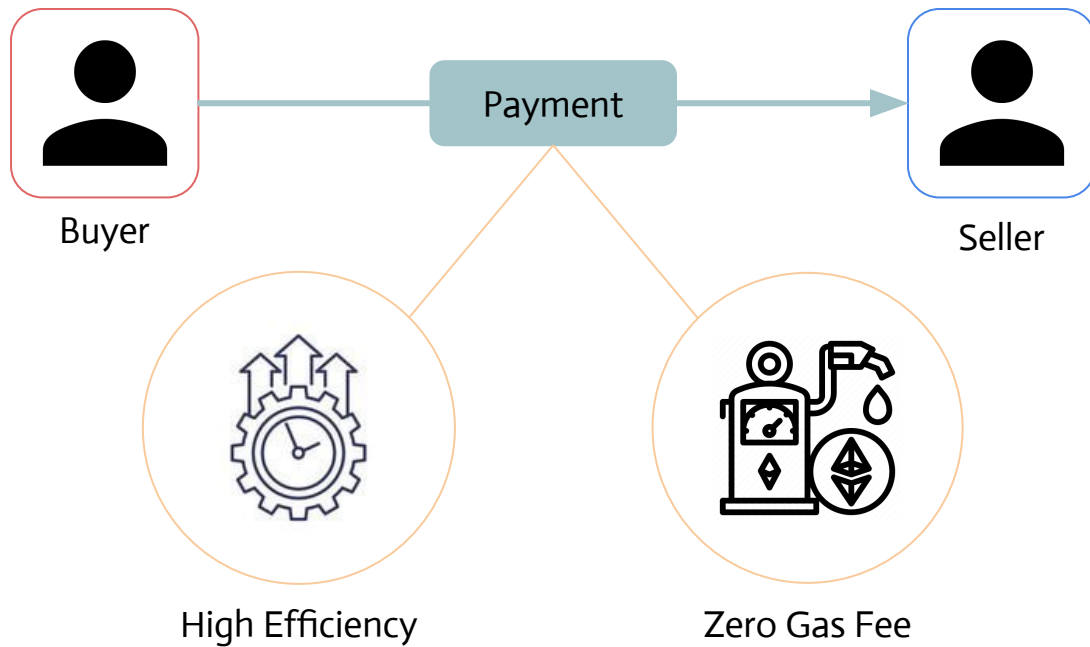

Batching Mechanism: Without Batching



Batching Mechanism: With Batching



Batching + Meta-Transaction

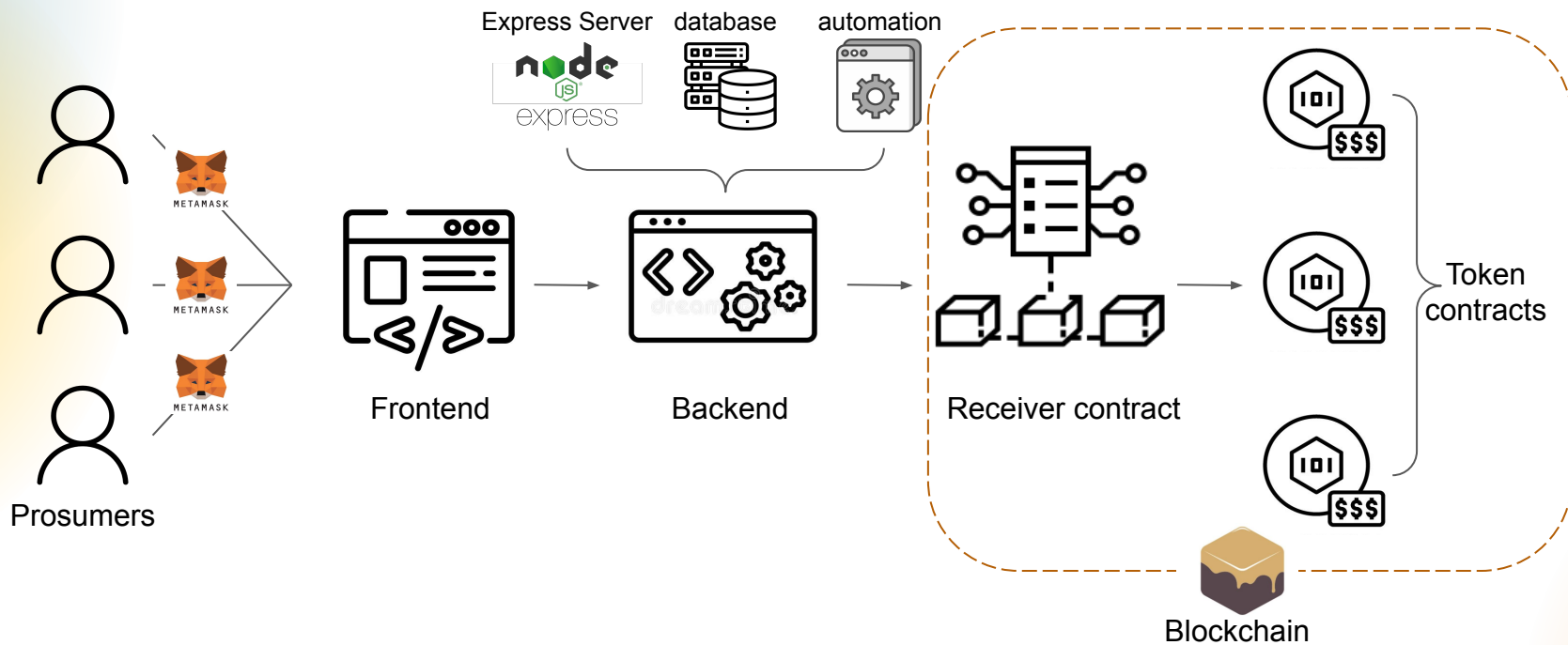


04

Implementation

```
25
26 def check_db():
27     if not os.path.isfile(FILE_URI):
28         db.create_all()
29
30 @app.route("/")
31 def home():
32     check_db()
33     all_books = db.session.query(Book).all()
34     return render_template("index.html", books=all_books)
35
36 @app.route("/edit", methods=['GET', 'POST'])
37 def edit():
38
39     if request.method == 'POST':
40         book_id = request.form["id"]
41         book_to_update = Book.query.get(book_id)
42         book_to_update.rating = request.form["rating"]
43         db.session.commit()
44         return redirect(url_for('home'))
```

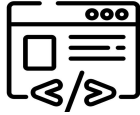
System Architecture



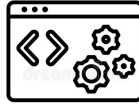
System Workflow



Prosumers



Frontend



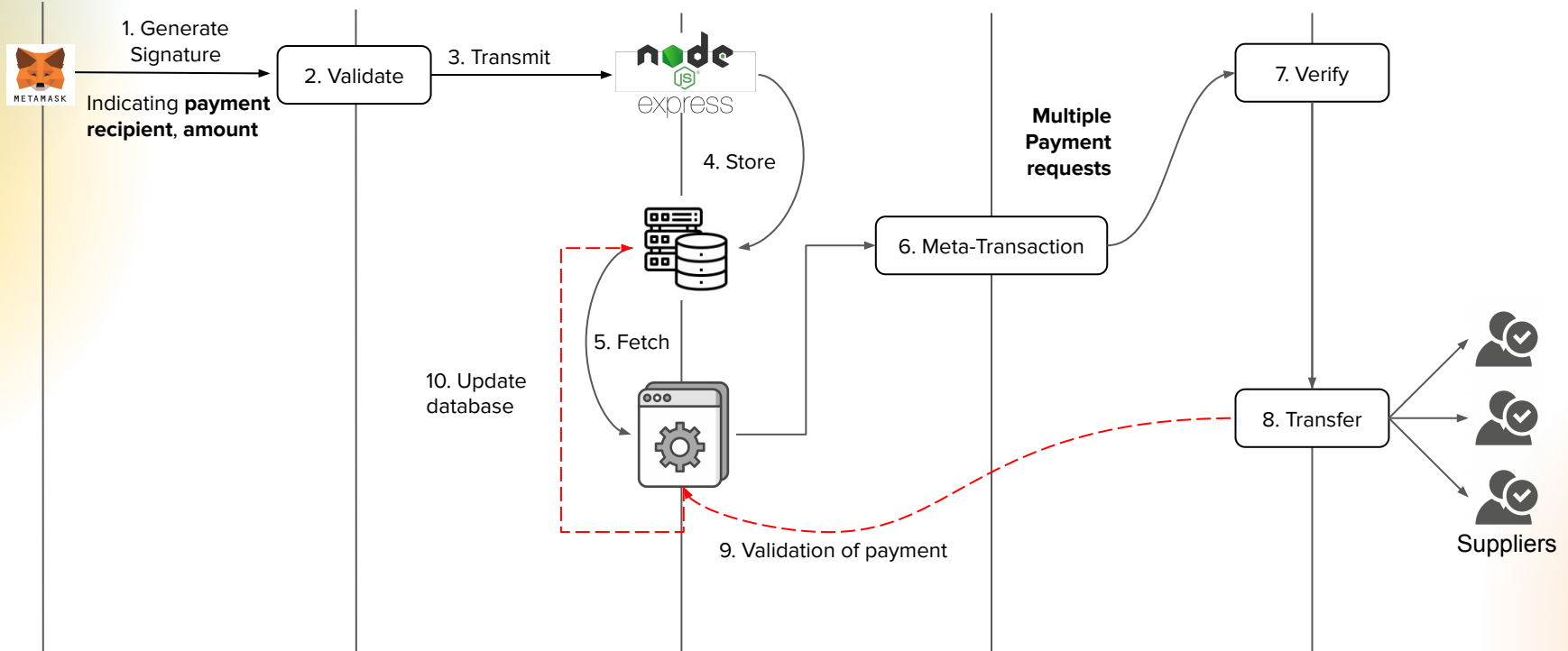
Backend



Receiver Contract



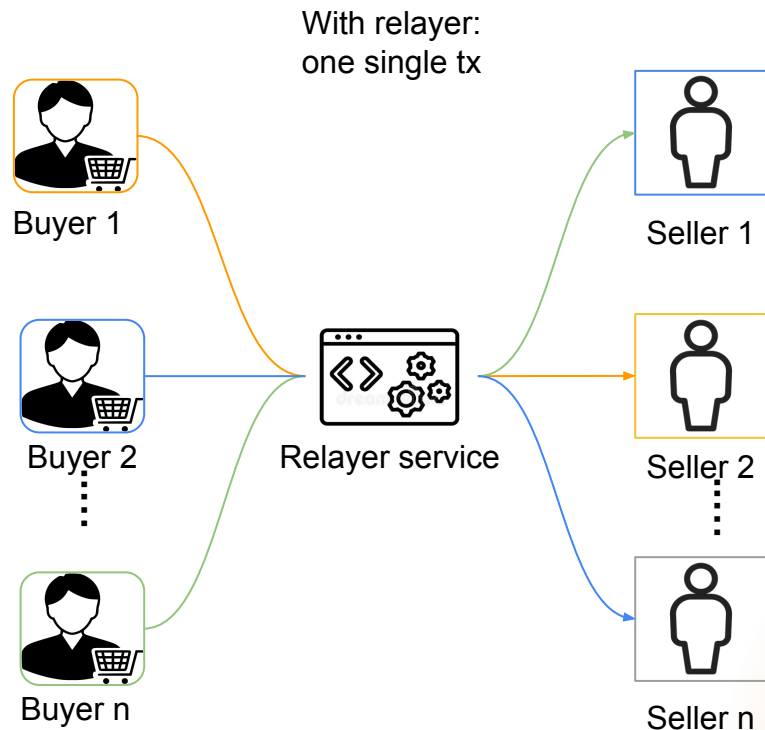
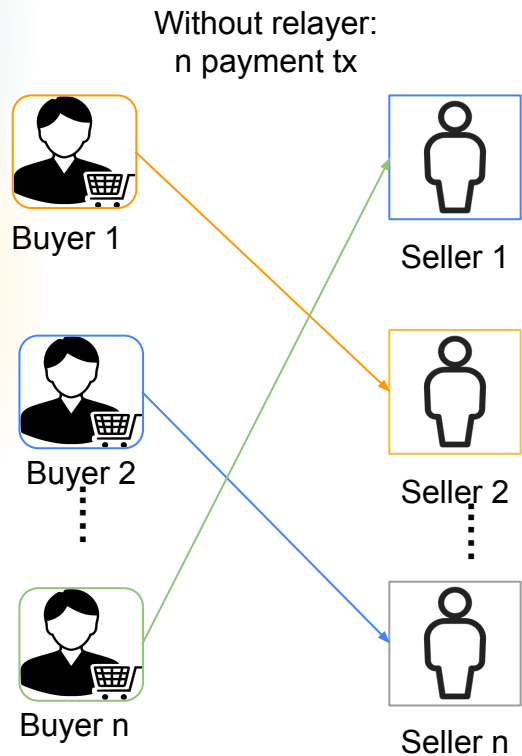
Payment Token



Demo



Performance Review: Efficiency



Performance Review: Usability



Prosumer

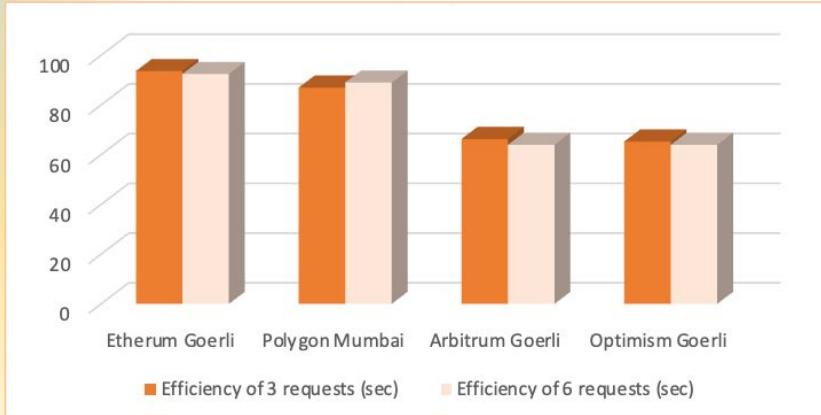
- Zero gas fee
- Do not need to hold native token in wallet



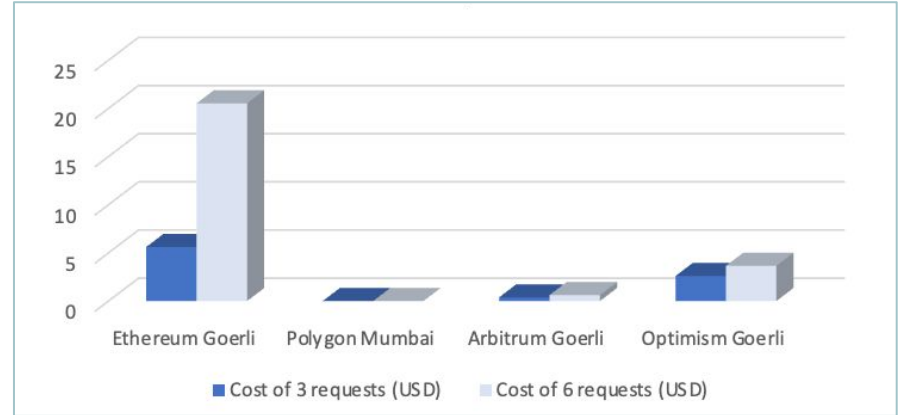
Relayer

- Pay gas fee for prosumer
- More payment requests being batched, less gas fee / request

Performance Review: Different Blockchain



Efficiency Comparison



Cost Comparison



05

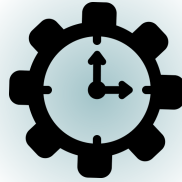
Conclusion

Summary

This project built a full-stack relay service that can fulfill prosumers' payment requests with high efficiency & improved usability



Relayer Service



High Efficiency



Improved Usability

Future Works

These future works can be implemented to further improve the relay service:



Decentralized Storage



Relayer Network



Economic Model

Thank you!

Q&A